Scalability and Future Plan

Field	Details
Date	02 November 2026
Team ID	N/A (Individual Project)
Project Name	EV Value & Longevity Analytics Dashboard
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Maximum Marks	1 Mark

The current version is a static, descriptive dashboard. The phases below show how it can grow without redesigning the data model.

Phase	Title	Description
Phase 1	Current State: Descriptive Dashboard	Static CSV files in a Tableau extract, nine worksheets, one dashboard and a five-chapter story on Tableau Public.
Phase 2	Sales Forecasting with Train/Test Split	Use the 200,000-row training file to fit a model (for example gradient boosting) predicting annual sales units from price, range, category, region and year. Evaluate on the 100,000-row test file with MAE and R-squared, then show predictions in Tableau.
Phase 3	Live Data and Automation	Move from CSV to a database (PostgreSQL or similar) with scheduled refresh so new model years appear automatically.
Phase 4	Total Cost of Ownership Calculator	Add insurance, warranty, resale value and charging data to build a per-vehicle cost-of-ownership view, plus charging-network and incentive data by region.

Technical Considerations

- Missing values (about 1% to 3% in five columns) need a documented treatment before modelling.
- Annual sales are right-skewed (median about 3,950, maximum above one million), so a log transform is suggested.
- Row counts above one million would need a database connection instead of a file.